Power



File: power.kicad_sch

Driver



File: driver.kicad_sch

MCU



File: mcu.kicad_sch

Connectors

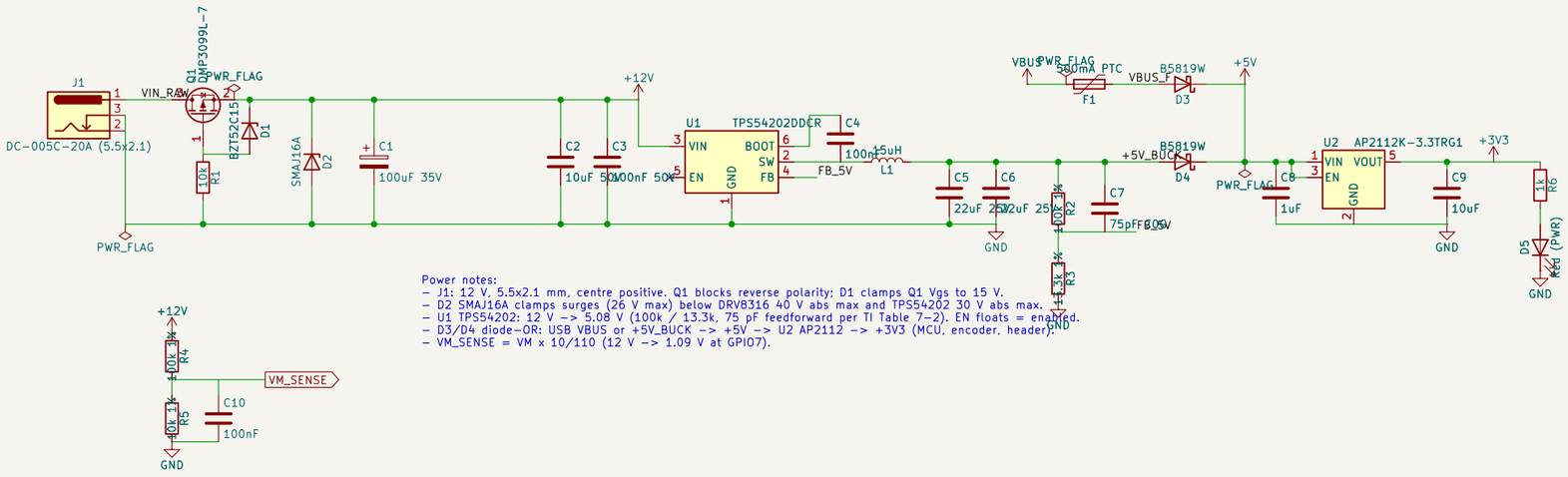


File: connectors.kicad_sch

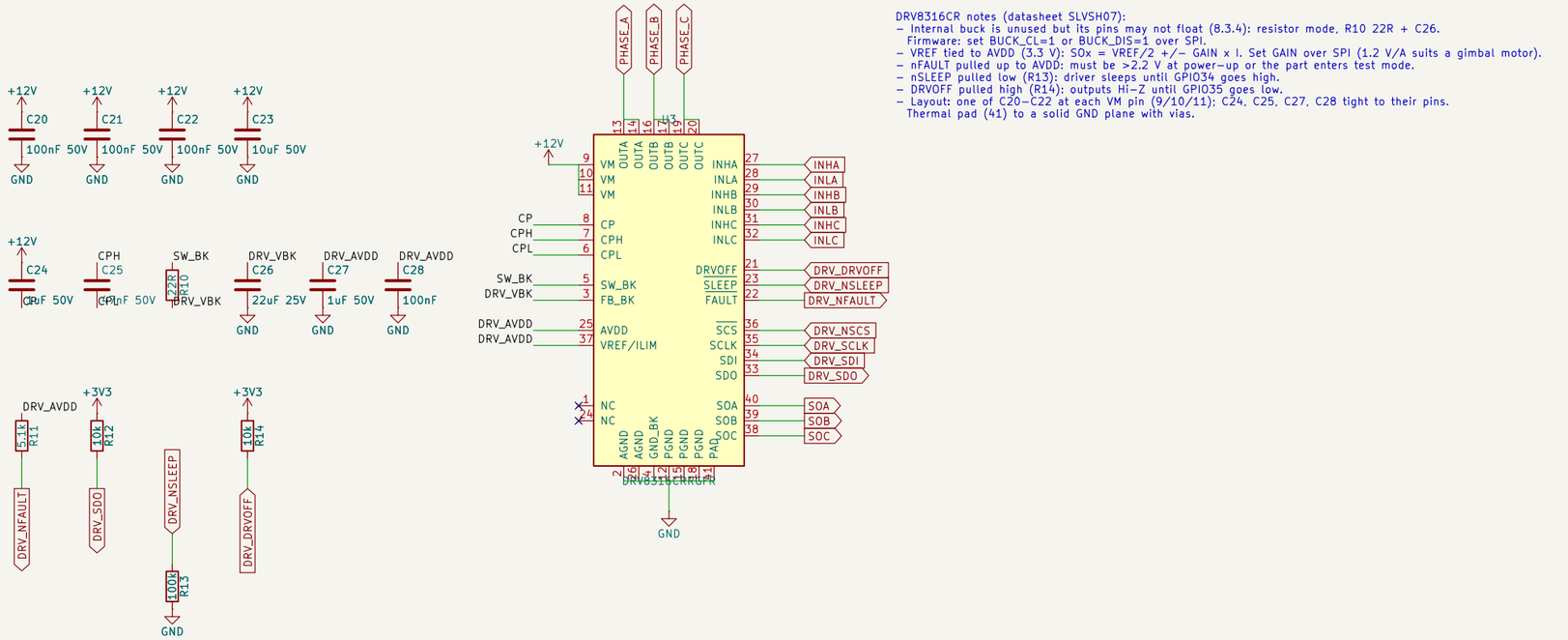
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File: smartknob_driver.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/5

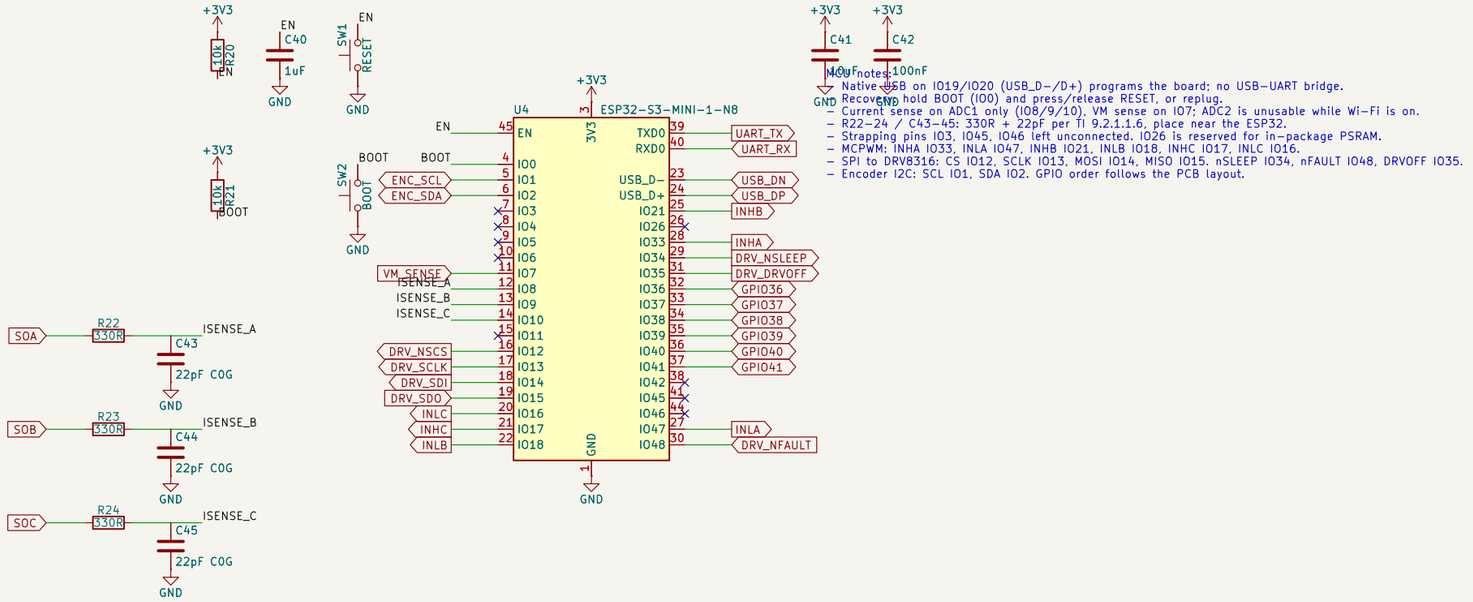


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KiCad E.D.A. 10.0.6		Id: 2/5

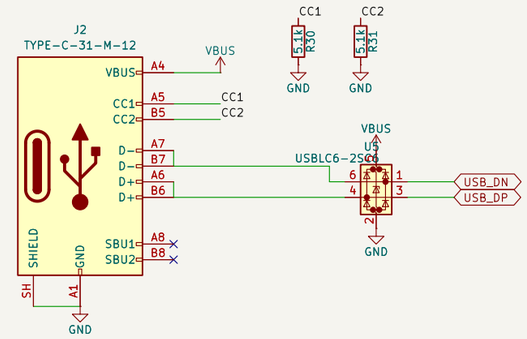


DRV8316CR notes (datasheet SLVSH07):
- Internal buck is unused but its pins may not float (8.3.4): resistor mode, R10 22R + C26.
  Firmware: set BUCK_CL=1 or BUCK_DIS=1 over SPI.
- VREF tied to AVDD (3.3 V): SOx = VREF/2 +/- GAIN x I. Set GAIN over SPI (1.2 V/A suits a gimbal motor).
- nFAULT pulled up to AVDD: must be >2.2 V at power-up or the part enters test mode.
- nSLEEP pulled low (R13): driver sleeps until GPIO34 goes high.
- DRVOFF pulled high (R14): outputs Hi-Z until GPIO35 goes low.
- Layout: one of C20-C22 at each VM pin (9/10/11); C24, C25, C27, C28 tight to their pins.
  Thermal pad (41) to a solid GND plane with vias.

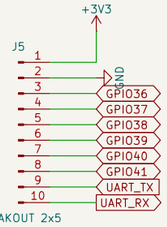
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Size: A3	Date:	**Rev:**
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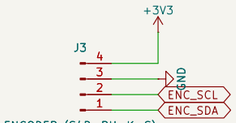


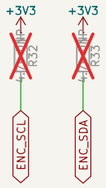
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KiCad E.D.A. 10.0.6		Id: 4/5

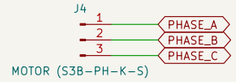












Connector notes:
- J3 ENCODER (JST-PH 4): 1 SDA, 2 SCL, 3 GND, 4 +3V3 (matches the AS5600 board cable). Check the cable keeps pin 1 on pin 1.
  R32/R33 are do-not-populate: the AS5600 board has its own pull-ups.
- J4 MOTOR (JST-PH 3): phases A/B/C. Wire order only changes direction; firmware calibrates it.
  Silkscreen: A B C / W R K.
- J5 BREAKOUT (2x5): 1 +3V3, 2 GND, 3 IO36, 4 IO37, 5 IO38, 6 IO39, 7 IO40, 8 IO41, 9 TX0, 10 RX0.
- H1–H4: M3 (3.2 mm) on a centred 24 mm square, not connected to GND.

Sheet: /Connectors/		
File: connectors.kicad_sch		
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Size: A3	Date:	**Rev:**
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